



MERN STACK DOCUMENTATION

BookStore Setup Guide

A field guide to scaffolding, configuring, and running the Client / Server architecture from a blank folder to a live app.

React

Vite

Node.js

Express

MongoDB

JWT

PREPARED FOR

SkillWallet — Full-Stack Development Program

Client / Server architecture reference & command log

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1 Project Overview

The **BookStore Management System** is a full-stack web application built on the MERN stack (MongoDB, Express.js, React.js, Node.js). The project lives inside a single parent directory named **BookStore**, which cleanly separates the client-facing interface from the server-side logic — keeping the codebase modular, easier to maintain, and simple to scale as new features are added.

Core Application Features

- Interactive 3D floating book design on the landing page
- Custom cursor animations for an engaging user experience
- Glassmorphism-based user interface styling
- Book browsing, filtering, and search functionality
- Shopping cart management for customers
- Secure user authentication and authorization pages
- Fully responsive design across desktop, tablet, and mobile devices

2 Project Folder Structure

A single root folder named **BookStore** contains two independent sub-projects — **Client** and **Server** — each with its own dependencies, configuration, and run scripts.

```
BookStore/  
  ■■■ Client/           React + Vite frontend application  
  ■   ■■■ src/  
  ■   ■■■ public/  
  ■   ■■■ package.json  
  ■■■ Server/           Node.js + Express + MongoDB backend  
    ■■■ models/  
    ■■■ controllers/  
    ■■■ routes/  
    ■■■ server.js  
    ■■■ .env
```

This structure keeps the user interface decoupled from server-side logic, making the project easier to manage, test, and deploy independently.

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Step-by-Step Setup Instructions

3.1 Create the Project Directory

Create a new root folder for the project, then create two subfolders inside it — one for the client and one for the server. Open the root folder in Visual Studio Code once both subfolders exist.

- Create a new folder and name it **BookStore**
- Inside it, create a folder named **Client**
- Inside it, create a folder named **Server**
- Open the **BookStore** folder in VS Code

3.2 Client Setup (React + Vite)

Open the **Client** folder in the VS Code integrated terminal and scaffold a new React application using Vite.

```
TERMINAL · /Client  
  
npm create vite@latest . -- --template react
```

When prompted, select the following options:

```
CLI PROMPT  
  
Select a framework:  
> React  
  
Select a variant:  
> JavaScript
```

Navigate into the generated client directory, install dependencies, and start the development server:

```
TERMINAL · /Client  
  
cd client  
npm install  
npm run dev
```

Once running, the frontend is accessible at:

```
CLIENT URL  
  
http://localhost:5173
```

3.3 Server Setup (Node.js, Express.js & MongoDB)

Open the **Server** folder in a separate VS Code terminal and initialize a Node.js project:

```
TERMINAL · /Server  
npm init -y
```

Create the following file and folders to organize backend logic:

Item	Type	Purpose
<code>server.js</code>	File	Application entry point; starts the Express server
<code>models/</code>	Folder	Mongoose schemas for books, users, and orders
<code>controllers/</code>	Folder	Business logic that handles requests and responses
<code>routes/</code>	Folder	Express route definitions mapping URLs to controllers

The backend manages REST API requests, JWT-based authentication and authorization, database operations for books, users and orders, core business logic, and secure communication between the frontend and the database.

Environment Configuration

Create a `.env` file in the root of the **Server** folder to store sensitive configuration values outside of version control:

```
.ENV · /Server/.env  
  
PORT=5000  
MONGO_URI=mongodb://localhost:27017/bookstore  
JWT_SECRET=bookstore_super_secret_jwt_key_2026  
JWT_EXPIRE=30d  
NODE_ENV=development
```

Install all backend dependencies, seed the database with sample data, and start the development server:

```
TERMINAL · /Server  
  
npm install  
npm run seed  
npm run dev
```

Running `npm run seed` populates the database with:

- 16 Indian and English books
- One customer account
- One administrator account
- A set of sample orders

Once running, the backend API is accessible at:

SERVER URL

`http://localhost:5000`

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Test Credentials

The seeded database includes two predefined accounts for testing customer-facing and administrative functionality.

Role	Email	Password
Customer	customer@bookstore.com	password123
Administrator	admin@bookstore.com	admin123

These credentials are intended strictly for local development and testing. Replace them with strong, unique values before deploying to a staging or production environment.

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Quick Command Reference

Action	Location	Command
Scaffold React app	Client/	<code>npm create vite@latest . -- --template react</code>
Install client deps	Client/	<code>npm install</code>
Run client dev server	Client/	<code>npm run dev</code>
Initialize backend	Server/	<code>npm init -y</code>
Install server deps	Server/	<code>npm install</code>
Seed database	Server/	<code>npm run seed</code>
Run server dev mode	Server/	<code>npm run dev</code>

End of document — BookStore Management System Setup & Configuration Guide